

Module-3

Evolution of Cellular Technologies: First Generation Cellular Systems, 2G Digital cellular systems – GSM and its Evolution, 3G Broadband Wireless Systems, Key Enabling Technologies and features of LTE, LTE Network Architecture.

Frequency Domain Multiple Accesses: Multiple Access for OFDM Systems, Orthogonal Frequency Division Multiple Access, Single Carrier Frequency Division Multiple Access.

[Text2: 1.2.1, 1.2.1.1, 1.2.2, 1.2.2.1, 1.2.3 (Only the mentioned sections and subsections), 1.4, 1.5, 4.1, 4.2, 4.3]

1. Explain 1G cellular system. 10 marks
 2. Explain 2G cellular system. 10 marks
 3. Explain 3G cellular system. 10 marks
 4. Explain 4G cellular system. 10 marks
 5. Explain 5G cellular system. 10 marks
 6. Write differences between 1G, 2G, 3G, 4G and 5G. 10 marks
 7. Explain GSM system Architecture. 10 marks
 8. Explain Key enabling technologies and features of LTE.
 9. Explain LTE Network architecture. 10 marks
 10. Explain the working of OFDM. 10 marks
 11. Explain the working of SC-FDMA. 10 marks
 12. Highlight the advantages and disadvantages of OFDM and SCFDMA. 10 marks
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13. Note on 3G Broad band wireless system. 10 marks
 14. Explain the evolution of cellular technologies 10 marks
Explain the evolution of wireless network. 10 marks
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1. Explain 1G cellular system. 10 marks

1G Wireless Communication (First Generation)**Introduction**

- 1G refers to the **first generation of wireless cellular technology**, introduced in the **early 1980s**.
- It was based on **analog transmission**.

Key Features

- Technology: **Analog voice communication**
- Frequency Bands: Mostly **800–900 MHz**
- Channel Access: **FDMA (Frequency Division Multiple Access)**
- Data Transmission: **Voice only** (no data, no SMS, no internet)
- Modulation: **Frequency Modulation (FM)**

Major Systems:

- NTT (Japan): First commercial cellular system (1979).
- NMT-400 (Europe): First to support international roaming (1981).
- AMPS (US) and its variants (ETACS/NTACS in Europe & Japan): The most successful systems. They were radio-similar but differed in channel bandwidth (AMPS: 30kHz, ETACS: 25kHz, NTACS: 12.5kHz).

Advantages

- Allowed **mobile voice communication** for the first time
- Wide coverage area

Limitations

- **Poor voice quality** due to noise and interference
 - **No security** – calls could be easily intercepted
 - **Low capacity** (limited number of users)
 - **No data services** (only analog voice)
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2. Explain 2G cellular system. 10 marks**2G (Second Generation Wireless Communication)**

- **Introduced:** Early 1990s
- **Technology:** Digital (unlike 1G which was analog)
- **Access Methods:** GSM (TDMA), CDMA
- **Data Rate:** Up to ~64 kbps
- **Frequency Band:** 850 MHz, 900 MHz, 1800 MHz, 1900 MHz

Key Features

- Digital voice calls (clearer & secure than 1G analog).
- **SMS (Short Message Service)** introduced for the first time.
- Basic data services (like MMS, fax, email).
- Better call quality & reduced noise.
- Improved spectral efficiency compared to 1G.
- Enabled **roaming** between countries using GSM standards.

Major 2G Systems:

- **GSM (Global System for Mobile Communications):** The most widely deployed 2G system.

- **IS-95 (CDMA):** Deployed in North America and parts of Asia.
- **IS-136 (TDMA):** A digital evolution of AMPS; deployed in North America but later discontinued and replaced.
- **PHS (Personal Handyphone System):** A cordless phone system with handover capability, deployed in Asia. Often categorized as a 2G technology.

New Services and Applications:

- **Short Messaging Service (SMS):** A hugely successful service that became a popular conversational tool and was later used for news, alerts, payments, and more.
- **Low Data Rate Wireless Data:**

Initially supported **circuit-switched data** (similar to dial-up).

Later evolved to support **packet data**.

Early services included news, stock quotes, and weather.

Technologies like **WAP (Wireless Access Protocol)** were developed to deliver Internet content to the limited displays of early mobile devices.

Advantages

- Secure voice communication (encryption used).
- Support for SMS & simple mobile internet (GPRS, EDGE upgrades).
- Energy-efficient compared to 1G.

Limitations

- Low data speed (not suitable for modern internet browsing).
- Mostly limited to **voice + text**, not multimedia.

3. Explain 3G cellular system. 10 marks

3G (Third Generation Wireless Communication)

- **Introduced:** Early 2000s
- **Technology:** Digital (based on **W-CDMA, CDMA2000, UMTS**)
- **Data Rate:**
 - Up to **2 Mbps** (stationary/indoor users)
 - ~384 kbps (mobile users)
- **Frequency Bands:** 1.8 – 2.5 GHz

Key Features

- **High-speed internet access** compared to 2G.

- **Video calling & conferencing** became possible.
- Support for **mobile TV, online gaming, multimedia messaging (MMS)**.
- **Always-on internet** via packet switching.
- Seamless global roaming.
- Enhanced security and encryption.

Advantages

- Faster data rates → enabled **web browsing, streaming, mobile apps**.
- Better voice quality & multimedia support.
- Foundation for **smartphones & mobile internet revolution**.

Limitations

- Higher cost for infrastructure and mobile devices.
- More battery consumption compared to 2G.
- Slower than 4G/5G, not suitable for heavy HD streaming.

Major 3G Systems:

GPRS (General Packet Radio Service):

- Introduced **packet-switched data**, a more efficient method for internet traffic.
- Added new core network elements: the **SGSN** (mobility management) and **GGSN** (connection to the internet).
- Theoretically offered speeds up to 160 kbps, with typical user rates of 20-40 kbps.

EDGE (Enhanced Data rates for GSM Evolution):

- Boosted data rates by introducing more efficient **8PSK modulation**.
- Provided a significant speed increase, with typical user rates of 80-120 kbps.
- This evolutionary path (GSM → GPRS → EDGE) formed the foundational architecture for later 3G systems.

4. Explain 4G cellular system. 10 marks

4G (Fourth Generation Wireless Communication)

- **Introduced:** Late 2000s – 2010s
- **Technology:** **All-IP (Internet Protocol)** based, using **LTE (Long-Term Evolution)** and **WiMAX**.

- **Data Rate:**
 - Up to **100 Mbps** (high mobility, e.g., cars/trains)
 - Up to **1 Gbps** (low mobility, e.g., stationary users)
- **Frequency Bands:** Widely varies (700 MHz – 2.6 GHz and beyond).

Key Features

- **High-speed broadband internet** (suitable for HD video streaming).
- **OFDM (Orthogonal Frequency Division Multiplexing)** for efficient spectrum use.
- **MIMO (Multiple Input Multiple Output)** antennas for faster data and better reliability.
- **Packet-switched network only** (no circuit switching).
- Support for **VoIP (Voice over IP)** → enabled apps like WhatsApp, Skype.
- Enhanced **QoS (Quality of Service)** for multimedia applications.

Advantages

- Extremely **high data rates** compared to 3G.
- Supports **HD video streaming, online gaming, video conferencing**.
- Global roaming with LTE networks.
- Low latency (~50 ms) → better user experience.

Limitations

- Requires costly **infrastructure upgrades**.
 - Higher **battery consumption** in early devices.
 - Coverage gaps in rural/remote areas (initially)
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5. Explain 5G cellular system. 10 marks

5G (Fifth Generation Wireless Communication)

- **Introduced:** Around **2019 onwards**
 - **Technology:** Advanced **all-IP network**, using **OFDM, Massive MIMO, Beamforming, mmWave, Network Slicing**
 - **Data Rate:**
 - Peak: up to **10–20 Gbps**
 - Average: **100 Mbps – 1 Gbps** (depending on deployment)
 - **Frequency Bands:**
 - **Low-band:** < 1 GHz (wide coverage, lower speed)
 - **Mid-band:** 1 – 6 GHz (balanced coverage & speed)
 - **High-band (mmWave):** 24 – 100 GHz (ultra-fast, short range)
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Key Features

- **Ultra-high speed** internet & downloads (10x faster than 4G).
- **Ultra-low latency (~1 ms)** → supports real-time applications.

- **Massive device connectivity** (up to **1 million devices/km²**) → essential for **IoT, smart cities**.
- **Network slicing** → multiple virtual networks on the same physical infrastructure.
- **Enhanced mobile broadband (eMBB)**, **Ultra-reliable low-latency communication (URLLC)**, and **Massive machine-type communication (mMTC)**.
- **Energy efficiency & better spectral efficiency**.

Advantages

- Enables **autonomous vehicles, telemedicine (remote surgery), smart factories, AR/VR, 8K streaming**.
- High **capacity & reliability** for dense urban areas.
- **Seamless connectivity** for IoT and AI-based applications.

Limitations

- **High deployment cost** (new towers, infrastructure upgrades).
- **Limited coverage** in rural areas
- Requires **compatible 5G devices**.
- Higher **security and privacy challenges** due to massive IoT connections.

6. Write Differences between 1G, 2G, 3G, 4G and 5G GSM

Generation	Period	Technology	Data Rate	Key Features	Limitations
1G	1980s	Analog (FDMA)	~2.4 Kbps (voice only)	• Voice calls (analog) • First mobile phones	• Poor voice quality • No data services • Low capacity & security
2G	1990s	Digital (TDMA, CDMA, GSM)	64 Kbps – 144 Kbps	• Digital voice (clearer) • SMS, MMS • Low-speed internet	• Slow data • Limited multimedia • Basic internet
3G	2000s	WCDMA, CDMA2000, UMTS	384 Kbps – 2 Mbps	• Mobile internet • Video calls • Online gaming • Multimedia support	• High cost • Limited speed compared to demand

Generation	Period	Technology	Data Rate	Key Features	Limitations
4G	2010s	LTE, OFDM, All-IP	100 Mbps – 1 Gbps	• High-speed broadband • HD video streaming • VoIP, Online gaming • Better security	• Expensive spectrum • Needs strong signal
5G	2020s	OFDM, Massive MIMO, Beamforming, mmWave	1–20 Gbps	• Ultra-high speed • Ultra-low latency (~1ms) • IoT & Smart cities • AR/VR, Autonomous vehicles • Network slicing	• Very high cost • Coverage issues (esp. mmWave) • New device requirement

7. Explain GSM network Architecture. 10 marks

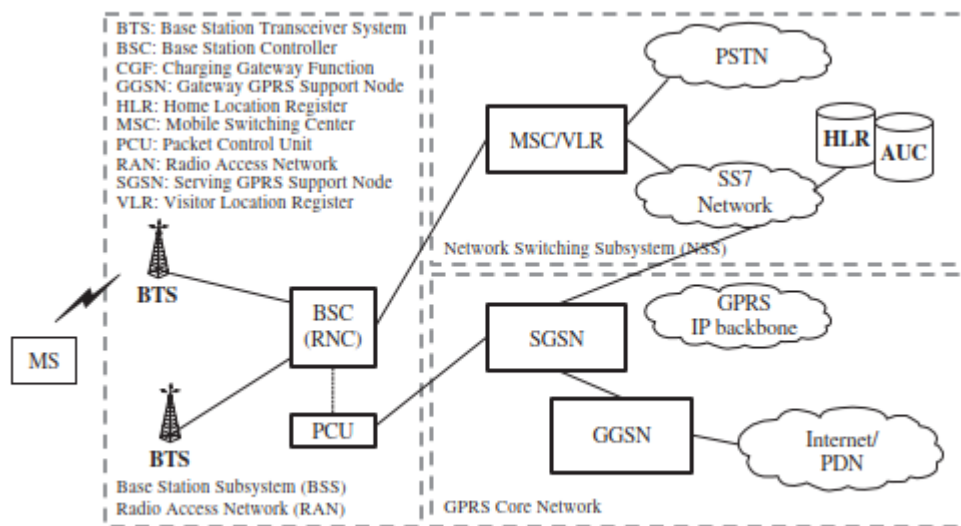


Figure 1.2 GSM network architecture.

Mobile Station (MS)

- User equipment (the phone + SIM card).
- Components:

Mobile Equipment (ME): Handset, modem, antenna.

SIM (Subscriber Identity Module): Stores IMSI, authentication keys, user data.

Base Station Subsystem (BSS)

Responsible for **radio communication** with mobile users.

- **BTS (Base Transceiver Station):**

Handles radio signals, modulation/demodulation.

Covers a cell area.

- **BSC (Base Station Controller):**

Manages multiple BTSs.

Handles handovers between BTSs, power control, frequency hopping.

MSC (Mobile Switching Center):

Call Routing & Switching → Connects mobile-to-mobile and mobile-to-PSTN calls.

Mobility Management → Handles handovers when a user moves between cells.

Location Management → Works with HLR and VLR to track subscriber location.

Billing & Charging → Collects call data records for billing.

Interfacing with Other Networks → Connects to PSTN, ISDN, internet, and other MSCs.

Security Functions → Authentication of subscribers and encryption control.

Roaming Support → Enables subscribers to make/receive calls while outside their home network.

HLR (Home Location Register):

Static data base

Stores permanent subscriber data (IMSI, MSISDN, subscription details).

Authentication support with AuC (security & encryption keys).

Location management (keeps track of subscriber's current VLR/MSC).

Call & SMS routing (provides routing info for incoming calls/SMS).

Roaming support (shares subscriber info with visited networks).

Service profile management (call forwarding, barring, data, roaming permissions).

VLR (Visitor Location Register):

Dynamic data base

Subscriber Data Storage (Temporary) → Stores details like IMSI, MSISDN, service subscription, and current location.

Location Updating → Updates when a subscriber moves to a new location area.

Call Management → Provides necessary info to MSC for call setup and routing.

Roaming Support → Maintains data for subscribers roaming in the MSC's area.

Authentication Support → Works with HLR/AuC for subscriber authentication.

Service Management → Keeps info about supplementary services (call forwarding, barring, etc.).

AuC (Authentication Center):

Subscriber Authentication → Verifies that the SIM card/user is genuine by using secret keys (Ki) and algorithms.

Encryption Key Generation → Generates ciphering keys (Kc) to encrypt voice and data.

Security Against Fraud → Prevents unauthorized network access.

Works with HLR → Stores authentication info linked to subscriber's permanent data in HLR.

Roaming Security → Provides authentication data to visited networks when the subscriber roams.

Packet control unit (PCU):

Allocates radio resources for packet data (GPRS/EDGE).

Schedules packets for multiple users (QoS management).

Segments & reassembles data into radio blocks.

Handles mobility for packet-switched users (cell changes).

Buffers packets when resources are busy.

Interfaces with SGSN through Gb interface.

Network sub system (NSS):

The NSS is the core part of GSM architecture, responsible for call switching, subscriber data, mobility, and interconnection with other networks.

Call switching and routing.

Location management (HLR + VLR).

Subscriber authentication and security (AuC).

Mobility and handover management.

Interfacing with PSTN/ISDN and other networks.

Serving GPRS Support Node (SGSN)

The SGSN is part of the GPRS core network, similar to MSC but for packet-switched data.

Delivers data packets between mobile users and the network.

Mobility management for data users (location updates, handovers).

Session management (PDP context activation/deactivation).

Security (authentication, ciphering).

Collects charging/billing info for packet data usage.

Gateway GPRS Support Node (GGSN)

The GGSN acts as a gateway between the GPRS network and external packet data networks (e.g., the internet).

Interfaces GPRS with external IP networks (Internet, corporate LAN).

Allocates IP addresses to mobile devices.

Converts GPRS packets into IP packets (and vice versa).

Routes packets between SGSN and external networks.

Handles charging and billing for external data usage.

SS7 Network (Signalling System No. 7)

SS7 is a global standard signalling protocol used in telecommunication networks (PSTN) for exchanging control information required to set up and manage telephone calls and other services.

It is used in both GSM (mobile) and PSTN (fixed) networks.

8. Explain Key enabling technologies and features of LTE.

1. Why OFDM Replaced CDMA for 4G/LTE

3G (UMTS/CDMA2000) uses CDMA, which is excellent for low-data-rate applications like voice, providing high system capacity through user multiplexing.

For high-data-rate applications, CDMA becomes inefficient as it requires impractically large bandwidths for effective spreading.

OFDM was chosen as the core modulation technology for LTE due to its superior performance in high-speed, broadband wireless scenarios.

2. Key Advantages of OFDM

Multipath interference: OFDM divides a high-speed data stream into many parallel, slower streams on orthogonal subcarriers. This increases the symbol duration, making the system resilient to inter-symbol interference (ISI) caused by multipath delays. A guard interval (cyclic prefix) is added to each symbol, which, if made longer than the delay spread, eliminates ISI completely.

Reduced Computational Complexity: OFDM can be efficiently implemented using FFT/IFFT algorithms, which simplifies receiver design, reduces cost, and lowers power consumption in mobile devices.

Graceful Performance Degradation: Unlike single-carrier systems that can experience abrupt performance drops, OFDM performance degrades gracefully. This means it can adapt to poor channel conditions by using more robust coding and modulation schemes, making it well-suited for adaptive modulation and coding.

Exploitation of Frequency Diversity: Coding and interleaving across subcarriers provide robustness against deep fades in specific frequency bands. It also allows for scalable channel bandwidths.

Efficient Multi-Access (OFDMA): When used as a multiple-access scheme, known as OFDMA, OFDM allows different subcarriers to be assigned to different users. This provides fine-grained channel allocation and significant capacity improvements, particularly in channels with slow-time variations.

Robustness: It is relatively immune to narrowband interference, as such interference affects only a small subset of subcarriers.

Suitable for Coherent Demodulation: OFDM facilitates easy pilot-based channel estimation, enabling power-efficient coherent demodulation.

Enables MIMO: OFDM converts a frequency-selective broadband channel into multiple flat-fading narrowband subchannels. This is the ideal environment for MIMO techniques to work effectively, significantly boosting system capacity.

Efficient Broadcast Support: Synchronized base stations can create a Single Frequency Network (SFN), where broadcast signals from multiple cells combine over the air to enhance signal power and coverage.

9. Explain LTE Network architecture. 10 marks

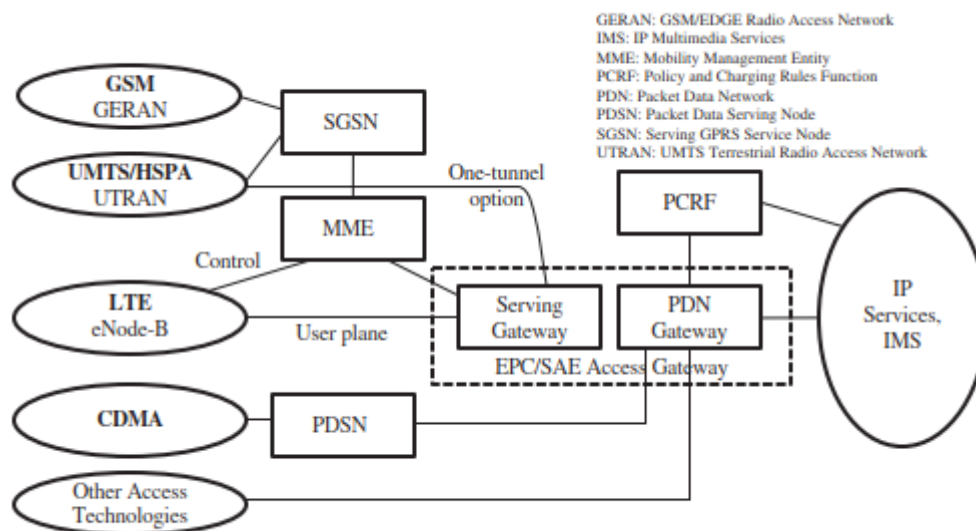


Figure 1.4 Evolved Packet Core architecture.

The LTE network architecture is centered around the **Evolved Packet Core (EPC)**, an all-IP core network designed for high capacity, low latency, and reduced cost. It supports advanced, media-rich services and can interwork with legacy 2G/3G networks.

The EPC consists of four main logical elements, each with a distinct function:

1. Mobility Management Entity (MME) - The Control Plane Manager

- This is the central brain for the control plane (signaling).
- **Key Functions:** Manages user access, authentication, security (e.g., issuing temporary identities), idle-mode location tracking, paging, and roaming.
- **Mobility:** Selects the appropriate gateways and handles handover control to 2G/3G networks.
- **Scale:** A single MME can manage thousands of eNode-Bs (the LTE base stations), a key difference from previous generations.

2. Serving Gateway (SGW) - The User Plane Mobility Anchor

- This gateway routes and forwards the user's actual data traffic.
- **Key Functions:** Acts as the mobility anchor for the user plane, handling handovers between eNode-Bs and between 3GPP technologies (LTE, 3G, 2G).
- **Other Roles:** Performs packet buffering, lawful interception, and provides data for accounting and charging.

3. Packet Data Network Gateway (PGW) - The Gateway to the Internet

- This is the connection point between the LTE network and the outside world of packet data networks (e.g., the internet).
- **Key Functions:** Allocates the user's IP address, performs packet filtering (including deep packet inspection), and enforces policy (e.g., data rate and QoS controls).
- **Anchor Point:** Serves as the stable anchor for the user's session towards external networks.

4. Policy and Charging Rules Function (PCRF) - The Policy & Charging Controller

- This node governs the quality of service (QoS) and charging rules.
- **Key Functions:** Makes real-time decisions on policy (e.g., which services get priority) and flow-based charging.
- **Scope:** It is mandatory for LTE and also supports non-3GPP access, like Wi-Fi.

In essence, the EPC creates a streamlined, flat architecture where:

- The **MME** handles all the signalling and control.
- The **SGW** and **PGW** handle the data path, with the SGW managing mobility and the PGW managing connectivity to the internet.
- The **PCRF** ensures the network resources are allocated according to the operator's policies.

10. Explain the working of OFDM and SCFDMA. 10 marks

Working of OFDMA

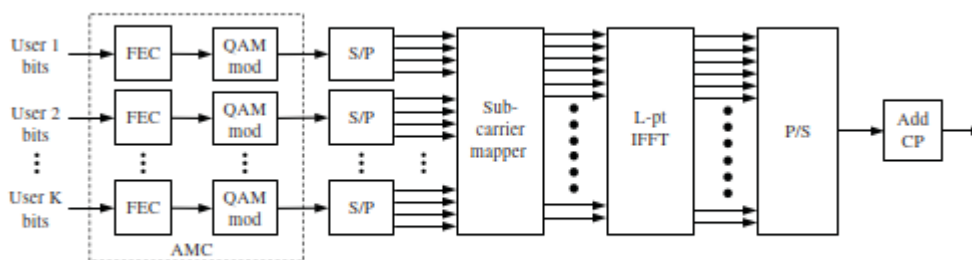


Figure 4.3 OFDMA downlink transmitter.

The fundamental purpose of the downlink transmitter is the opposite of the uplink: **it multiplexes data for multiple users into a single signal that is broadcast** from the base station. Each user's data is placed on a unique set of subcarriers within the same OFDM symbol.

1. User Bits

- Each user (User 1, User 2 ... User K) has its own input data stream.

2. FEC (Forward Error Correction)

- Adds redundancy to user bits to protect against channel errors.

- Typical methods: Convolutional coding, Turbo coding, or LDPC.
3. **QAM Modulator (Quadrature Amplitude Modulation)**
 - Converts the binary bits into complex symbols suitable for transmission.
 - QAM can be 16-QAM, 64-QAM, etc., depending on channel conditions.
 4. **S/P (Serial to Parallel Conversion)**
 - Converts the serial stream of symbols into parallel streams for OFDM subcarriers.
 - Each parallel stream corresponds to a specific subcarrier.
 5. **Subcarrier Mapper**
 - Allocates subcarriers to different users according to the OFDMA scheduling.
 - Ensures users' signals are orthogonal (no interference).
 6. **L-point IFFT (Inverse Fast Fourier Transform)**
 - Converts the frequency-domain symbols (mapped to subcarriers) into a **time-domain OFDM signal**.
 - This allows all subcarriers to be transmitted simultaneously without interference.
 7. **P/S (Parallel to Serial Conversion)**
 - Converts the parallel time-domain samples back to a serial stream for transmission.
 8. **Add CP (Cyclic Prefix)**
 - Adds a **cyclic prefix** to each OFDM symbol to prevent **Inter-Symbol Interference (ISI)** caused by multipath propagation.
 - Ensures orthogonality between subcarriers is preserved in a multipath channel.

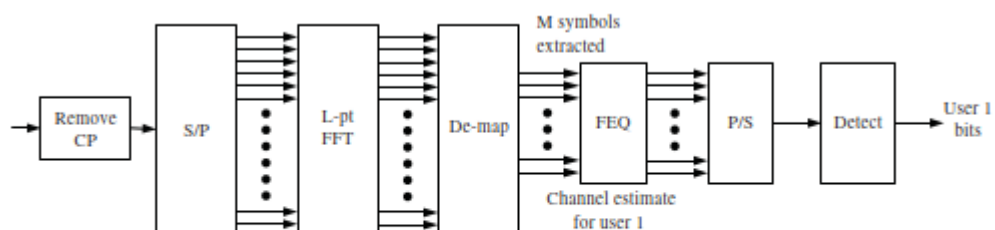


Figure 4.4 OFDMA downlink receiver for user 1. Each of the K active users—who by design have orthogonal subcarrier assignments—have a different receiver that only detects the M_k subcarriers intended for it.

The key function of OFDMA downlink receiver (mobile phone) is to extract only its own data from the complex composite signal broadcast by the base station, which contains data for many users simultaneously.

1. Remove CP (Cyclic Prefix)

- In OFDMA, each transmitted symbol includes a **Cyclic Prefix (CP)** to prevent **inter-symbol interference (ISI)** caused by multipath propagation.
- The receiver first **removes the CP** before further processing, recovering the original symbol duration.

2. S/P (Serial-to-Parallel Conversion)

- OFDMA uses multiple subcarriers simultaneously.
- The incoming serial stream is converted into **parallel streams**, one for each subcarrier, so the FFT can process them efficiently.

3. L-point FFT

- The FFT transforms the time-domain signal into the frequency domain.
- Each subcarrier corresponds to a **frequency bin**, separating the contributions of different users/subcarriers.
- **L** is the FFT size, which matches the number of total subcarriers in the system.

4. De-map

- After FFT, the receiver **extracts the subcarriers** allocated to this user (User 1).
- For OFDMA, each user has **orthogonal subcarriers**, so each receiver only focuses on its own set of subcarriers M_k

5. FEQ (Frequency-Domain Equalizer)

- The wireless channel introduces **fading and phase shifts** for each subcarrier.
- The **FEQ compensates for these effects** using the channel estimate for the user, recovering the transmitted symbols as accurately as possible.

6. P/S (Parallel-to-Serial Conversion)

- The parallel streams (one per subcarrier) are recombined into a **serial stream** of symbols.

7. Detect

- Finally, the receiver **demodulates the symbols** (e.g., QAM) into **bits** for User 1.

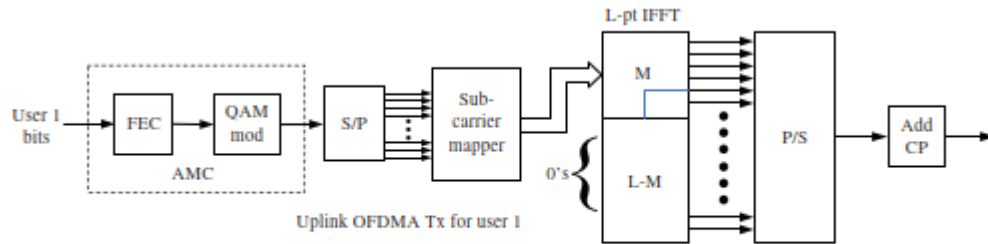


Figure 4.5 OFDMA uplink transmitter for user 1, where user 1 is allocated subcarriers 1, 2, ..., M of L total subcarriers.

The purpose OFDMA uplink transmitter is to convert a user's raw data bits into an OFDMA signal that can be transmitted over the radio channel, using only the specific subcarriers assigned to that user.

1. Input Data & FEC (Forward Error Correction):

- The user's raw data bits ("User 1 bits") first pass through a **FEC** encoder.
- This adds redundant bits (error-correcting code) to the data. This allows the receiver to detect and correct errors that occur during transmission, making the link more robust.

2. QAM Modulation (Quadrature Amplitude Modulation):

- The encoded bitstream is mapped into complex numbers (symbols) using a modulation scheme like 16-QAM or 64-QAM.
- This converts the digital bits into an analog waveform representation. Each symbol represents a unique combination of amplitude and phase, allowing multiple bits to be transmitted per symbol. The specific scheme is chosen by the **AMC (Adaptive Modulation and Coding)** block based on channel quality.

3. Serial-to-Parallel Conversion (S/P):

- The stream of QAM symbols is converted from a single serial stream into M parallel streams.
- This is a fundamental step for OFDMA. Instead of sending symbols one after another, they will now be sent simultaneously on different subcarriers.

4. Sub-carrier Mapper:

- This is a crucial step for OFDMA. The M parallel symbols from the user are placed onto the specific subcarriers (frequencies) assigned to **User 1** (in this case, subcarriers 1 through M). The remaining subcarriers ($L - M$ of them) are filled with zeros (0's).
- This ensures that User 1's signal only occupies its allocated portion of the total available bandwidth (L subcarriers), allowing other users to simultaneously use the other subcarriers without interference.

5. Inverse Fast Fourier Transform (IFFT):

- The entire block of L subcarriers (containing User 1's M symbols and many zeros) is processed by the IFFT.
- The IFFT mathematically converts the signal from the frequency domain (where we have individual subcarriers) into the time domain. The output is a single, complex time-domain waveform that is a summation of all the orthogonal subcarriers. The zeros ensure that User 1's energy is only present in its assigned subcarriers.

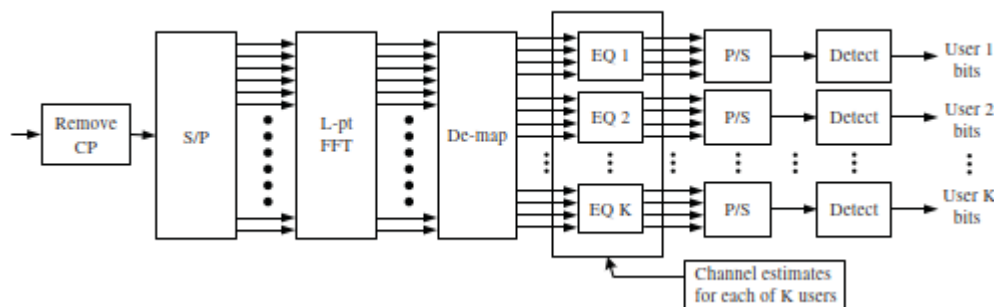
6. Parallel-to-Serial Conversion (P/S):

- The parallel time-domain samples output from the IFFT are converted back into a single, serial stream of samples.
- This creates a continuous digital representation of the OFDMA symbol that is ready for final transmission processing.

7. Add CP (Cyclic Prefix):

- A copy of the end of the OFDMA symbol is taken and appended to the beginning. This appended portion is the Cyclic Prefix.
- The CP acts as a guard interval. It helps to mitigate Inter-Symbol Interference (ISI) caused by multipath propagation (where signals reflect and arrive at the receiver at slightly different times). It also preserves the orthogonality of the subcarriers.

11. Explain the working of SC-FDMA. 10 marks



The BTS (uplink receiver) process the composite signal received from all users, extract the data from each user, and correct for distortions introduced by the radio channel.

1. Remove CP (Cyclic Prefix):

- The first step is to remove the Cyclic Prefix from the received time-domain signal.
- This discards the part of the signal that contains inter-symbol interference (ISI) caused by multipath delay. This step is crucial for maintaining orthogonality between the subcarriers.

2. Serial-to-Parallel (S/P):

- The long, serial stream of time-domain samples is converted into parallel blocks. Each block corresponds to the samples of one OFDMA symbol.
- To prepare the signal for the FFT, which processes a block of samples at a time.

3. FFT (Fast Fourier Transform):

- Each block of time-domain samples is passed through the FFT.
- The FFT converts the signal from the **time domain back to the frequency domain**. The output of the FFT is the raw data on all L individual subcarriers. This composite signal contains the data from all K users, superimposed on their respectively assigned subcarriers.

4. De-map:

- This is the critical multiple-access step. The receiver knows which subcarriers were assigned to which user (e.g., User 1: subcarriers 1-10, User 2: subcarriers 11-25, etc.).
- The de-mapper plucks out the symbols from the subcarriers assigned to each user. It routes User 1's symbols to User 1's processing chain, User 2's symbols to User 2's chain, and so on.

5. EQ 1, EQ 2, ... EQ K (Equalization - Per User):

- Each user's stream of symbols passes through a dedicated **Equalizer**.
- The radio channel distorts the signal, causing amplitude and phase shifts that are different for each subcarrier and for each user (since users are in different physical locations). The equalizer uses **channel estimates** (measurements of how the channel affected known pilot signals) to reverse this distortion. It corrects the amplitude and phase of each symbol so that it looks as close as possible to what was originally transmitted.

6. Parallel-to-Serial (P/S - Per User):

- The parallel stream of equalized symbols for each user is converted back into a serial stream.
- To prepare the symbol stream for the detection stage.

7. Detect (Detection / Demodulation - Per User):

- This block is the QAM demodulator. It takes each corrected complex symbol and decides which QAM constellation point (e.g., which combination of bits in 16-QAM) was most likely transmitted.
- To convert the symbols back into a stream of data bits.

8. Output:

- The final output is the recovered **User 1 bits, User 2 bits, ..., User K bits**.

12. Explain the working of SC-FDMA. 10 marks.

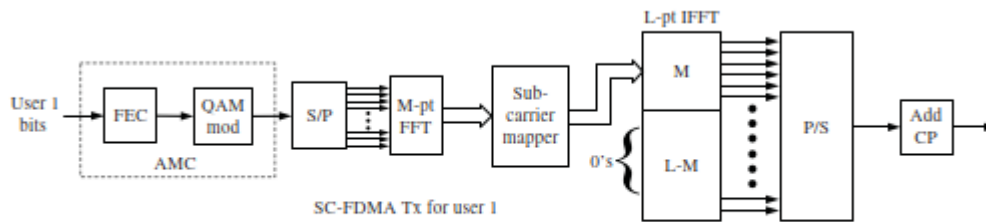


Figure 4.8 SC-FDMA uplink transmitter for user 1, where user 1 is allocated subcarriers 1, 2, ..., M of L total subcarriers.

SC-FDMA is the modulation scheme used in the LTE uplink (from the user's device to the base station).

- **Used** to combine the multi-user flexibility of OFDMA with the low **Peak-to-Average Power Ratio (PAPR)** of single-carrier systems.
- Low PAPR is crucial for mobile devices as it leads to higher power amplifier efficiency, longer battery life, and lower cost.

Transmitter (MS):

- **FFT (DFT) Precoding:** The user's block of complex symbols is first processed by a small FFT (or DFT). This converts the time-domain signal into the frequency domain *before* the main IFFT.
- **Subcarrier Mapping:** The output of this small FFT is then mapped onto the specific subcarriers allocated to that user (just like in OFDMA).
- **IFFT:** The signal is then processed by a large IFFT, as in standard OFDMA, to create the final time-domain signal for transmission.
- The final transmitted signal is a single-carrier signal that occupies only its assigned portion of the spectrum. This process inherently creates a low-PAPR signal.

Receiver (Base Trans-receiver Station):

- The receiver is very similar to an OFDMA receiver but includes an extra step to reverse the precoding.
- After the main FFT, equalization, and de-mapping, each user's signal passes through a small **IFFT**.
- This IFFT reverses the initial FFT precoding done by the transmitter, bringing the signal back to its original form for detection.

SC-FDMA is essentially **OFDMA with an added FFT precoding step** at the transmitter. This extra step sacrifices a small amount of processing complexity to achieve the major benefit of low PAPR, making it the ideal choice for the power-sensitive uplink in cellular systems like LTE.

12. Highlight the advantages and disadvantages of OFDM and SCFDMA

OFDMA: used for the Downlink (from the base station to phone).

Advantages	Disadvantages
High Spectral Efficiency: Excellent at packing data into available spectrum.	High PAR (Peak-to-Average Ratio): Requires expensive, power-hungry amplifiers.
Robust to Interference: Handles frequency-selective fading well.	Sensitive to Synchronization: Needs very precise timing and frequency alignment.
Simple Equalization: Easy to process at the receiver.	Spectral Leakage: Can cause more interference to adjacent channels.

SC-FDMA: used for the Uplink (from phone to the base station).

Advantages	Disadvantages
Low PAR (Peak-to-Average Ratio): The big one! Allows for cheaper, more battery-efficient power amplifiers in phones.	Lower Spectral Efficiency: Doesn't pack data as tightly as OFDM.
Better Battery Life: Due to low PAR, it's less draining on the mobile device's battery.	More Complex Transmitter/Receiver: Requires extra processing steps (an extra FFT) compared to OFDM.
Good Range: Lower PAR allows for a more constant and potentially stronger average signal.	Performance Trade-offs: Can experience more out-of-band interference (spectral leakage).